Date 8th septemember 2024.

ICT Assignment

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Q1). (1). $(256)_{10} \rightarrow (100000000)_2$

256	128	64	32	16	8	4	2	1
1	0	0	0	0	0	0	0	0

(2). $(1024)_{10} \rightarrow (10000000000)_2$

1024	512	256	128	64	32	16	8	4	2	1
1	0	0	0	0	0	0	0	0	0	0

(3). $(512)_{10} \rightarrow (1000000000)_2$

512	256	128	64	32	16	8	4	2	1
1	0	0	0	0	0	0	0	0	0

Q2). (1) $(345)_{10} \rightarrow (159)_{16}$

$345 \div 16 \rightarrow 21.5625$ 21 R 9

$21 \div 16 \rightarrow 1.3125$ 1 R 5

$1 \div 16 \rightarrow 0.0625$ 0 R 1

(2). $(789)_{10} \rightarrow (315)_{16}$

$789 \div 16 \rightarrow 49.3125$ 49 R 5

$49 \div 16 \rightarrow 3.0625$ 3 R 1

$3 \div 16 \rightarrow 0.1875$ 0 R 3

(3). $(2048)_{10} \rightarrow (800)_{16}$

$2048 \div 16 \rightarrow 128$ 128 R 0

$128 \div 16 \rightarrow 8$ 8 R 0

$8 \div 16 \rightarrow 0.5$ 0 R 8

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3). (1). $(512)_{10} \rightarrow (1000)_8$

$512 \div 8 \rightarrow 64$ 64R0

$64 \div 8 \rightarrow 8$ 8R0

$8 \div 8 \rightarrow 1$ 1R0

$1 \div 8 \rightarrow 0.125$ 0R1

(2). $(1234)_{10} \rightarrow (2322)_8$

$1234 \div 8 \rightarrow 154.25$ 154R2

$154 \div 8 \rightarrow 19.25$ 19R2

$19 \div 8 \rightarrow 2.375$ 2R3

$2 \div 8 \rightarrow 0.25$ 0R2

(3). $(999)_{10} \rightarrow (1747)_8$

$999 \div 8 \rightarrow 124.875$ 124R7

$124 \div 8 \rightarrow 15.5$ 15R4

$15 \div 8 \rightarrow 1.875$ 1R7

$1 \div 8 \rightarrow 0.125$ 0R1

4). (1). 2^8 101101011 $\rightarrow (363)_{10}$

2^6 2^5 2^3 2^1 2^0

$(2^8 \times 1) + (2^6 \times 1) + (2^5 \times 1) + (2^3 \times 1) + (2^1 \times 1) + (2^0 \times 1) = 363$

(2). 1101001010 $\rightarrow (842)_{10}$

2^9 2^8 2^6 2^3 2^1

(3). $(2^9 \times 1) + (2^8 \times 1) + (2^6 \times 1) + (2^3 \times 1) + (2^1 \times 1) = 842$

1111110000 $\rightarrow (1008)_{10}$.

2^9 2^8 2^7 2^6 2^5 2^4

$(1 \times 2^9) + (1 \times 2^8) + (1 \times 2^7) + (1 \times 2^6) + (1 \times 2^5) + (1 \times 2^4) = 1008$

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Q5). (1). $(1A3)_{16} \rightarrow (419)_{10}$

$16^2\ 16^1\ 16^0 = 1 \times 16^2 + 10 \times 16^1 + 3 \times 16^0 = 419$

(2). $(3F8)_{16} \rightarrow (1016)_{10}$

$16^2\ 16^1\ 16^0 = 3 \times 16^2 + 15 \times 16^1 + 8 \times 16^0 = 1016$

$(C9B)_{16} \rightarrow (3227)_{10}$

$16^2\ 16^1\ 16^0 = (12 \times 16^2) + 9 \times 16^1 + 11 \times 16^0 = 3227$

Q6). (1). $(457)_8 \rightarrow (303)_{10}$

$8^2\ 8^1\ 8^0 = 4 \times 8^2 + 5 \times 8^1 + 7 \times 8^0 = 303$

(2). $(1052)_8 \rightarrow (554)_{10}$

$8^3\ 8^2\ 8^1\ 8^0 = 1 \times 8^3 + 0 \times 8^2 + 5 \times 8^1 + 2 \times 8^0 = 554$

(3). $(764)_8 \rightarrow (500)_8$

$8^2\ 8^1\ 8^0 = (7 \times 8^2) + (6 \times 8^1) + (4 \times 8^0) = 500$

Q7). (1). $10011001101 \rightarrow (4CD)_{16}$

8421   8421   8421

0100   1100   1101

4   C   D

(2). $1101011011 \rightarrow (35B)_{16}$

21   8421   8421

0011   0101   1011

3   5   B

(3). $1010101010 \rightarrow (2AA)_{16}$.

0010   1010   1010

8421   8421   8421

2   A   A

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8). (1). $(2\ B5)_{16} \rightarrow (0010\ 1011\ 0101)_2$

2 11 5
8421 8421 8421
0010 1011 0101

(2). $(7\ D\ 3)_{16} \rightarrow (0111\ 1101\ 0011)_2$

7 D 3
8421 8421 8421
0111 1101 0011

(3). $(F\ A\ 4)_{16} \rightarrow (1111\ 1010\ 0100)_2$.

F A 4
8421 8421 8421
1111 1010 0100

9). (1). $(726)_8 \rightarrow (111\ 010\ 110)_2$

7 2 6
421 421 421
111 010 110

(2). $(453)_8 \rightarrow (100\ 101\ 011)_2$

4 5 3
421 421 421
100 101 011

(3). $(1075)_8 \rightarrow (001\ 000\ 111\ 101)_2$

421 421 421 421
001 000 111 101

10). (1). $(\underbrace{111}\underbrace{000}\underbrace{110})_2 \rightarrow (706)_8$

421 421
7 0 6

(2). $(1001100110)_2 \rightarrow (1146)_8$

001 1 421 421
421 1 421 421
1 1 4 6

(3). $(1011010111)_2 \rightarrow (1327)_8$.

421 421 421
1 3 2 7